Hoggatt sums fail infinite log-convexity for every $m \geq 4$

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Abstract

We disprove Conjecture 4.3 of Fang–Zhang–Zhao in its current arXiv formulation. For parameter $m = 4$ or 5, the fourth log-convexity iterate has a negative entry. For every $m \geq 6$, the third iterate already has a negative entry. The separate Baxter case $m = 3$ is not decided. The identified journal final has not been read; campaign eligibility remains pending.

Task 1. Full disproof and uniform extension

Use exactly the source's convention

$$(\mathcal{L}a)_n = a_n a_{n+2} - a_{n+1}^2.$$

Infinite log-convexity requires every entry of $\mathcal{L}^r a$ to be nonnegative for every integer $r \geq 1$. Define the Hoggatt sums from [1, p. 3] by

$$A_n(m) = H_n^{[m]}(1,1) = \sum_{k=0}^{n-1} \frac{\prod_{i=0}^{m-1} \binom{n+m-2}{k+i}}{\prod_{j=1}^{m-1} \binom{n+m-2}{j}}, \qquad n \geq 1. \tag{1}$$

Conjecture 4.3 asserts infinite log-convexity for every fixed integer $m \geq 2$.

Theorem 1. *The sequence $(A_n(m))_{n\geq 1}$ is not infinitely log-convex for any integer $m \geq 4$. More precisely,*

$$\begin{aligned} (\mathcal{L}^4 A(4))_2 &= -1115094590484444626406114654006866031 36, \\ (\mathcal{L}^4 A(5))_2 &= -4404568077343728457084971349790650891111078 2976, \end{aligned} \tag{2}$$

and $(\mathcal{L}^3 A(m))_2 < 0$ for every integer $m \geq 6$.

The finite certificate at $m = 4$

Substitution in (1) gives

$$(A_1(4), \ldots, A_{10}(4)) = (1, 2, 7, 32, 177, 1122, 7898, 60398, 494078, 4274228). \tag{3}$$

These are the known terms of OEIS A005362, with index shifted by one; their numerical discovery is not claimed here. Applying the displayed definition of $\mathcal{L}$ gives

n	$(\mathcal{L}A(4))_n$	$(\mathcal{L}^2A(4))_n$
1	3	420
2	15	22400
3	215	8967705
4	4575	5312555556
5	139062	6332625182576
6	5388152	10985704681554720
7	254309640	
8	14041752660	

(4)

The next row is

n	$(\mathcal{L}^3A(4))_n$
1	3264676100
2	38581511487375
3	28565867977326238944
4	1826002473957302112002854 4

(5)

Consequently $(\mathcal{L}^4A(4))_2$ equals

$$38581511487375 \cdot 1826002473957302112002854 4 - 28565867977326238944^2,$$

which is the negative integer in (2). This already disproves the stated universal conjecture. The entry uses only A_2 through A_{10}, so there is no choice of a boundary value involved.

For $m = 5$, the ten initial values from (1) are

$$(1, 2, 8, 44, 310, 2606, 25202, 272582, 3233738, 41454272). \tag{6}$$

Four applications of the same subtraction give the second value in (2). The complete intermediate rows for both parameters are included in the exact verification output.

A polynomial certificate for every $m \geq 6$

Writing $\ell = n - 1 - k$, factorial cancellation in (1) gives

$$A_n(m) = \sum_{k=0}^{n-1} M(k, n-1-k; m), \qquad M(k, \ell; m) = \prod_{i=1}^{k} \prod_{j=1}^{\ell} \frac{m+i+j-1}{i+j-1}. \tag{7}$$

For example, both expressions for each summand equal $\prod_{i=0}^{m-1}(k+\ell+i)!i!/((k+i)!(\ell+i)!)$. Empty products are 1. Formula (7) expresses every $A_n(m)$ as a polynomial in m with rational coefficients.

Set

$$Q(m) = m^4 + 9m^3 + 26m^2 + 36m + 72$$

and

$$\begin{aligned} P(m) = {} & m^{17} + 42m^{16} + 723m^{15} + 6462m^{14} + 27742m^{13} \\ & - 16148m^{12} - 934842m^{11} - 5962420m^{10} - 21758691m^9 \\ & - 53818158m^8 - 97778329m^7 - 138523978m^6 \\ & - 159475276m^5 - 154920392m^4 - 129902448m^3 \\ & - 88784352m^2 - 37081152m - 2892672. \end{aligned} \tag{8}$$

Expanding (7) for $2 \le n \le 8$ and applying $\mathcal{L}$ three times yields the polynomial identity

$$(\mathcal{L}^3 A(m))_2 = -\frac{m^4(m-1)^4(m+1)^5(m+2)^2 Q(m)P(m)}{1547934105600}. \tag{9}$$

This is a finite algebraic identity, directly reproducible from (7). The verifier checks it by exact polynomial arithmetic and independently by 37 evaluations of the source binomial sums; the degree is at most 36.

It remains to prove the sign uniformly. The function

$$\frac{P(m)}{m^{13}} = m^4 + 42m^3 + 723m^2 + 6462m + 27742 - \sum_{j=1}^{13} c_j m^{-j}$$

has $c_j > 0$ by (8), so its derivative is strictly positive for $m > 0$. Direct evaluation gives

$$P(6) = 264264378826752 > 0.$$

It follows that $P(m) > 0$ for every real $m \ge 6$. All other factors in (9) are positive there, proving the negative sign and completing the theorem.

Verification, scope and attribution

The verifier uses exact integer and rational arithmetic throughout. It compares the two defining products, reproduces both finite certificates, checks the degree-36 identity in two ways, and supplies an additional sign certificate: all 18 coefficients of $P(z+6)$ are strictly positive. The infinite family follows from (9) and the proved sign, not from numerical extrapolation. Review is by the authoring agent only.

The source already establishes ordinary log-convexity at $m = 4$; the negative fourth iterate does not contradict that weaker result. The Catalan case $m = 2$ is known to be infinitely log-convex. Conjecture 4.2, the Baxter case $m = 3$, remains separate. The row and column log-concavity results in [3, 4] concern different sequences. The complete current preprint [1] was read. Its identified journal final, DOI 10.1016/j.jmaa.2026.130615, remains unread, so this package receives zero new campaign credit pending the final-source audit.

Tasks 2–5. Follow-up directions

2. Decide infinite log-convexity for the remaining Baxter parameter $m = 3$.
3. Determine the largest r for which the entire Hoggatt sequence is r-log-convex, as a function of $m \ge 4$.
4. Prove eventual r-log-convexity for every fixed m, r, with explicit bounds on the starting index.
5. Locate all negative entries of the third and fourth iterates, and determine how their positions depend on m.

References

[1] H. Fang, C. X. T. Zhang and J. J. Y. Zhao, *Analytic properties arising from the Baxter numbers*, arXiv:2505.05873v1, current 14-page preprint; Conjecture 4.3 on p. 11.

[2] OEIS Foundation, A005362, Hoggatt sequence with parameter 4. Entry by N. J. A. Sloane and S. Plouffe.

[3] J. Mao and W. Shi, *The analytic properties of Hoggatt triangles*, arXiv:2607.01582v1, 2026.

[4] Y. Liu and J. Mao, *Pfaffian–Toeplitz identities, Schur positivity, and the q-log-convexity of Baxter polynomials*, arXiv:2608.22258v1, 2026.

[5] MathDB, Problem 368345, accessed September 18, 2026.